

Rutile & Ilmenite

UK resource prospects, global supply and procurement pathways

Prepared for	Hydrogrid PLC
Purpose	Internal screening of potential raw-material sources for rutile and ilmenite, including UK prospectivity and international suppliers.
As at	10 October 2026
Status	Preliminary market-intelligence memo - not a reserve report, bankable feasibility study, or supplier offer.

Executive conclusion

Hydrogrid should treat imported feedstock as the practical near-term route. The UK has geological indications of titanium-bearing heavy minerals offshore, but the British Geological Survey reports no known significant accumulations of rutile or ilmenite on the UK Continental Shelf and no current offshore metallic-mineral workings. Supplier discussions should start with Eramet Grande Cote (Senegal), Kenmare/Moma (Mozambique), Iluka (Australia), Tronox (Australia/South Africa) and IREL (India), subject to technical qualification and confirmation of uncommitted merchant volumes. [1-8]

Scope assumption. This memo treats rutile and ilmenite as mineral feedstocks. Finished rutile titanium dioxide pigment is a different product and is subject to a separate UK trade-remedy investigation; see section 3. No annual requirement, end-use process, or product specification was supplied, so the shortlist is a route to qualification rather than a validated procurement decision.

Prepared using publicly available geological, government, and company disclosures. Figures may use different reporting bases; the units are identified where material.

1. Decision brief

The evidence supports a two-track strategy: pursue commercial supply from established international producers now, while treating UK geological work as a low-cost optionality exercise rather than a source of planned production.

Decision area	Assessment	Implication for Hydrogrid
UK primary mine supply	Low near-term prospectivity. BGS says significant UKCS rutile/ilmenite accumulations are not known; some offshore heavy-mineral sands are prospective but poorly defined. [1]	Do not build a supply plan or capital case around domestic extraction without new sampling, resource work and permits.
Ilmenite supply	Broadest market and higher-volume feedstock. USGS estimated 9.4 Mt of global 2025 mine production on a contained-TiO ₂ basis. [2]	Seek at least two qualified suppliers and price by delivered cost per contained tonne of TiO ₂ , adjusted for impurities and process route.
Natural rutile supply	Smaller market: USGS estimated about 450 kt global mine production in 2025 on a contained-TiO ₂ basis. Australia, Sierra Leone and South Africa are important producers. [2,3]	Qualify multiple sources and avoid treating rutile as interchangeable across all applications.
Best initial suppliers	Eramet Grande Cote, Kenmare/Moma, Iluka, Tronox and IREL offer credible product pathways. [4-8]	Issue common RFQs and request assays, supply calendars, trace-element data, origin and delivery terms.
Trade-policy watch	On 7 October 2026 the UK TRA proposed a measure on rutile TiO ₂ pigment from China. This is not the same as raw rutile mineral concentrate. [9]	Classify the exact material and customs code before landed-cost modelling; do not transfer pigment duties to ore by assumption.

Recommended sourcing posture

- Base-load ilmenite: test Eramet Grande Cote and Kenmare/Moma in parallel; use IREL and/or a qualified integrated producer as a third route. Kenmare is a large producer, but its H1 2026 ilmenite output fell 39% year on year, while rutile fell 44%, making current allocation and recovery assumptions particularly important. [5]
- Natural rutile: request grade-specific offers from Eramet Grande Cote, Iluka and Tronox. Confirm merchant availability rather than relying on mine-level production statistics; vertically integrated producers may reserve feedstock for their own pigment plants. [4,6,7]
- UK optionality: commission a desk study and data request focused on offshore heavy-mineral sands only if domestic resilience has strategic value. Do not infer an economic resource from mineral occurrence records. [1]

Critical technical gate

First confirm the actual downstream process. Ilmenite (commonly around 45-65% TiO₂) usually requires upgrading for some applications; natural rutile is typically around 92-96% TiO₂; synthetic rutile and titanium slag are alternative upgraded feedstocks. Price, recoverability, trace impurities and process compatibility can matter more than quoted tonnes alone. [7,8]

2. United Kingdom: source potential

2.1 Geological picture

The main public UK evidence is cautionary. The British Geological Survey (BGS) states that significant accumulations of rutile and ilmenite are not known on the UK Continental Shelf (UKCS). It identifies offshore areas of north-east Scotland and the Sea of the Hebrides where heavy-mineral sands have been recorded and where titanium minerals are likely to occur, but it also notes that offshore metallic-mineral data are sparse and resource properties poorly known. There are currently no workings for metallic minerals on the UKCS. [1]

Area / evidence	What is known	Commercial interpretation
North-east Scotland offshore	BGS identifies offshore heavy-mineral sands; rutile/ilmenite may be present in the mineral assemblage. [1]	Exploration lead only. No published bankable resource, grade continuity, recoveries or mining plan established by the cited evidence.
Sea of the Hebrides / nearshore western Scotland	BGS notes heavy-mineral-sand areas and considers titanium minerals likely to occur in some sediments. [1]	Potential for sampling and mineralogical work, but environmental sensitivity, marine access, sediment dynamics and recovery costs are major constraints.
Northumbrian beach / nearshore sands	Older BGS regional reporting describes rutile and zircon in beach sands in garnet-rich bands; offshore heavy-mineral concentrations have also been reported. [10]	Historical occurrence evidence, not proof of an economically mineable deposit. Coastal protection and sediment-management rules would be central.
Onshore mineral occurrences	Mineral specimen or accessory-mineral occurrences, including rutile in some rocks, can be recorded without forming a mineable orebody.	Do not count as a commercial source absent systematic bulk sampling, grade/tonnage definition and metallurgical tests.

2.2 UK pathway and screening work

- Near term (0-3 months): obtain BGS geological and seabed datasets; review existing sample locations, grain-size data and heavy-mineral assays; request an expert view on likely titanium-mineral assemblages and sampling gaps.
- Exploration stage (3-18 months): only if initial data justify it, conduct permitted sampling and mineralogical assay. Establish total heavy minerals (THM), valuable heavy minerals (VHM), rutile/ilmenite shares, size distribution and liberation; model recovery and tailings/water impacts.
- Development stage: would require resource definition, environmental assessment, stakeholder engagement, marine and mineral permissions, port/logistics design, rehabilitation and an independently reviewed economic case. These steps are not currently demonstrated by public UK resource data.
- Alternative UK lever: look for potential recovery from existing mineral-processing residues or industrial waste streams, but treat this as a separate recycling/by-product programme requiring its own material-flow evidence and assay work.

UK conclusion

UK exploration may add long-term optionality, but the evidence does not support presenting UK rutile or ilmenite as an immediately available domestic resource. For supply continuity, imported offtake, inventory buffers and product qualification are currently more realistic levers. [1,11]

Policy context: the UK Government's Vision 2035 Critical Minerals Strategy emphasises resilience through midstream capability, recycling, domestic production where viable and diversified international partnerships. This supports a diversified sourcing approach but does not itself establish UK rutile/ilmenite reserves. [11]

3. Global market and supply geography

USGS 2026 estimates for 2025 production provide a useful country-level benchmark. The table below reports thousand metric tonnes of contained TiO₂, not gross tonnes of shipped mineral concentrate. Mine-product tonnage will be higher or lower depending on grade and product form; the figures should therefore not be used directly as supplier allocation or purchase volumes. [2]

3.1 Ilmenite - leading 2025 producers

Country	2025 production	Indicative position
China	3,200 kt	Largest producer; sizeable domestic consumption and potential geopolitical/trade-policy exposure.
Mozambique	1,900 kt	Second-largest producer; major merchant-scale source through Moma/Kenmare.
South Africa	1,300 kt	Large established producer; supply tied to mineral-sands operators and upgrading routes.
Australia	780 kt	Large, technically mature market with natural rutile and synthetic-rutile capabilities.
Norway	390 kt	Established ilmenite output; check exact product, availability and downstream commitments.
Canada	360 kt	Ilmenite-bearing resources and downstream upgrading; may be more suitable for upgraded feedstock routes.
Senegal	370 kt	Established coastal mineral-sands route; Eramet Grande Cote markets multiple titanium products.
India	240 kt	Domestic producer via IREL; confirm export eligibility and product grade.

3.2 Rutile - leading 2025 producers

Country	2025 production	Procurement relevance
Australia	200 kt	Largest rutile producer; Iluka and Tronox are relevant named supplier candidates.
Sierra Leone	110 kt	Material rutile-producing jurisdiction; check current operating and export status directly before qualifying a seller.
South Africa	100 kt	Established output, including Tronox/Richards Bay Minerals supply chains.
India	13 kt	IREL markets rutile; smaller global volume than ilmenite.
Mozambique	10 kt	Moma has rutile as a by-product of much larger ilmenite output.

USGS reported rounded world totals of approximately 9.4 million tonnes for ilmenite and 450,000 tonnes for rutile in 2025, measured on a contained-TiO₂ basis. Ilmenite accounts for roughly 90% of titanium-mineral consumption, but rutile is materially less abundant and generally a higher-grade natural product. [2]

How to read this market

Mine production is not equivalent to open merchant supply. Some producers integrate feedstock into pigment or smelting, while others sell mineral products directly. A procurement assessment must validate saleable product form, contract allocations, grade and delivered logistics with each producer.

4. Supplier shortlist and sourcing routes

The candidates below are practical leads for initial qualification, not confirmed offers. Priorities reflect product match and evidence of commercial production; availability, price, credit terms and exact technical fit remain unverified.

Supplier / country	Materials and evidence	Key issue to check	Priority
Eramet Grande Cote Operations Senegal	Markets ilmenite grades 54 and 58, a standardised ilmenite 56, natural rutile and leucoxene. Published product data list ilmenite at 54-58% TiO ₂ , rutile around 95% and leucoxene around 90%. [4]	Ask for current exportable volume, assay ranges, consistency, packing/loading options, lead time and the exact specification of the grade proposed. Rutile product is described as suitable for arc-welding applications; that is not automatic proof of suitability for titanium metal production.	A - first RFQ
Kenmare Resources / Moma Mozambique	Large-scale ilmenite producer with rutile and zircon co-products. Kenmare reported 2025 output of 842,300 t ilmenite and 8,600 t rutile. [5]	H1 2026 production declined year on year: ilmenite 273,100 t (-39%) and rutile 2,700 t (-44%). H1 shipments were supported by stock drawdown. Confirm 2026-27 allocations and plant transition schedule. [5]	A - volume, with supply caveat
Iluka Resources Australia	2025 production: 335.3 kt ilmenite, 78.9 kt natural rutile and 209.9 kt synthetic rutile. Balranald began producing heavy-mineral concentrate in 2025. [6]	Iluka suspended Cataby and synthetic-rutile kiln 2 from 1 Dec 2025 amid weak TiO ₂ pigment demand. Distinguish natural rutile / ilmenite merchant volumes from internal processing and suspended capacity. [6]	A/B - natural rutile option
Tronox Australia and South Africa	Integrated producer of ilmenite, natural rutile, leucoxene, synthetic rutile and titanium slag. Its 2025 SEC filing reports roughly 161.9 kt of natural rutile across listed operations. [7]	Vertically integrated into TiO ₂ pigment; merchant availability may be limited or product-specific. Request exact material form, TiO ₂ content, impurity data and allocation window.	B - qualified enquiry
IREL (India) Limited India	Government-owned company with mineral-separation operations and product lines including ilmenite and rutile. IREL has advertised spot-sale enquiries for ilmenite. [8]	Confirm grade, seller eligibility, export approvals, documentation, delivery basis and any controls associated with beach-sand minerals and co-produced radioactive minerals.	B - diversification
Richards Bay Minerals / Rio Tinto South Africa	Established heavy-mineral-sands operation producing titanium minerals and related products. Rio Tinto approved the Zulti South investment in March 2026 to extend long-term operating continuity. [12]	Treat current supply as a commercial enquiry; do not model future Zulti South capacity as immediate. Development is a long-term continuity measure, not an instant new source.	B - strategic enquiry

Recommended RFQ order

- Wave 1: Eramet Grande Cote, Kenmare and Iluka - one Senegalese source, one large Mozambique ilmenite source and one Australian rutile/synthetic-rutile source.
- Wave 2: Tronox and IREL - useful diversification, with a particular focus on merchant volume, export eligibility and end-use suitability.
- Watchlist: South African RBM supply/expansion; Sierra Leone rutile only after confirming current operations, seller authority and shipment track record.

Supplier data are from public disclosures and have not been independently audited for this memo. Company production volumes and USGS country output are different measures and should not be compared as if they were the same unit.

5. Product fit, risks and procurement plan

5.1 Define the product before comparing offers

Product form	Typical TiO ₂ content (indicative)	Procurement consideration
Ilmenite concentrate	About 45-65%	Often the high-volume, lower-cost feedstock; may need smelting to titanium slag or upgrading to synthetic rutile depending on process. Iron, gangue and trace elements drive value. [7]
Natural rutile	About 92-96%	Higher-grade natural feedstock; relatively constrained global output. Particle size, trace-element profile and intended process matter. [7]
Synthetic rutile	About 89-92%	Upgraded product made from ilmenite by reducing/leaching iron. Can be an alternative where the downstream process accepts it; availability depends on kiln capacity and operating decisions. [7]
Titanium slag	Often 86-89%	Smelted ilmenite product; not raw mineral concentrate. Confirm chloride/sulfate suitability and impurity tolerances before tendering. [7]
Rutile TiO₂ pigment	Pigment/preparation with $\geq 80\%$ by weight TiO ₂ under the relevant UK case definition	Chemical pigment, not natural rutile mineral. UK anti-dumping proceedings involving China should be assessed against the specific product and customs codes only. [9]

5.2 Quality and commercial diligence checklist

- Assays and mineralogy: total TiO₂; total iron/FeO; SiO₂, Al₂O₃, Cr₂O₃, V₂O₅, P, S and other process-specific impurities; moisture; particle-size distribution; mineral liberation; and radioactivity / U-Th where relevant to the deposit assemblage.
- Representative sample: obtain a certificate of analysis and sealed bulk sample from the actual saleable product; use independent laboratory testing and downstream bench trials. A headline TiO₂ percentage is not enough to establish suitability.
- Commercial terms: quote basis (FOB/CFR/CIF or delivered UK), port and bag/bulk format, annual and monthly committed tonnes, minimum lot, pricing formula, payment security, rejected-load rules, assay settlement, delivery windows and force-majeure protections.
- Responsible sourcing: request permits, mine-life outlook, rehabilitation and water-management plans, tailings controls, labour and community policies, grievance mechanisms and origin traceability. For any co-produced monazite or radioactive-bearing material, require legal and safety review appropriate to the actual product.
- Logistics: model ocean freight, port storage, demurrage, insurance, FX, inspection, handling losses and inland haulage to the consuming site. Compare delivered cost per contained tonne of TiO₂, then adjust for recovery, process yield and impurities.

5.3 30-day action plan

Timing	Action	Deliverable
Week 1	Confirm end-use route, acceptable product forms, annual volume, ramp profile and technical specification.	One-page purchase specification and volume forecast.
Week 1-2	Issue identical RFQs to Eramet Grande Cote, Kenmare, Iluka, Tronox and IREL. Request product data, samples, 12-24 month supply plans and incoterms.	Comparable supplier returns and qualified-volume register.
Week 2-3	Run independent assays on samples; ask process engineering to test recovery, slag/metal/pigment compatibility and contaminant rejection.	Technical pass/fail list and adjusted cost comparison.
Week 3-4	Validate customs codes, UK trade measures, seller/export eligibility, sanctions and origin; negotiate trial cargo or sample agreement with top two.	Shortlist of two primary sources plus one backup.

Decision gate for domestic UK work

Fund a UK seabed or recycling study only if public-data review and preliminary mineralogical evidence indicate sufficient recoverable rutile/ilmenite, manageable permitting and a plausible landed-cost or resilience advantage over imported feedstock. Otherwise, preserve capital for qualification, inventory and supplier diversification.

6. Sources and limitations

Sources below were reviewed on 10 October 2026. URLs are clickable. Company output is reported as stated by each source; USGS mine-production data are expressed on a contained-TiO₂ basis unless the cited source states otherwise.

- [1] British Geological Survey (2013), Other minerals, Mineral Resources of Scottish Waters and the Central North Sea, report OR/13/013. Used for UKCS prospectivity, evidence gaps and absence of workings. [Open source](#)
- [2] U.S. Geological Survey (2026), Mineral Commodity Summaries 2026 - Titanium Mineral Concentrates, estimates for 2025 mine production and reserves. [Open source](#)
- [3] Geoscience Australia (2025), Australian minerals industry review / commodity summaries: Mineral Sands, including 2024 Australian rutile and ilmenite resource and production context. [Open source](#)
- [4] Eramet Grande Cote, Products and Activities, product grades and commercial product description for ilmenite, rutile and leucoxene. [Open source](#)
- [5] Kenmare Resources, The Moma Mine and H1 2026 production / half-year report, including 2025 products and H1 2026 year-on-year production changes. [Open source](#)
- [6] Iluka Resources, 2025 Annual Report, product output and suspension of Cataby and synthetic-rutile kiln 2 from December 2025. [Open source](#)
- [7] Tronox, 2025 Form 10-K filed with the US SEC, product definitions, indicative TiO₂ grades, operating locations and 2025 natural-rutile output. [Open source](#)
- [8] IREL (India) Limited, company overview and product pages for ilmenite and rutile. [Open source](#)
- [9] UK Trade Remedies Authority, 7 October 2026, proposed anti-dumping measure on rutile titanium dioxide imports from China; active case AD0086. [Open source](#)
- [10] British Geological Survey, The geology of the central North Sea, regional report, including historical reports of rutile and zircon in Northumbrian beach sands. [Open source](#)
- [11] UK Department for Business and Trade, UK Critical Minerals Strategy / Vision 2035, published 22 November 2025, updated 23 January 2026. [Open source](#)
- [12] Rio Tinto, 2 March 2026, Rio Tinto approves Zulti South investment of \$473 million and lifts project suspension. [Open source](#)

Limitations

- This is a desktop sourcing screen, not a geological resource statement, independent technical due-diligence report, legal opinion or supplier quotation.
- Public company production is not proof of merchant availability. Volumes, prices and terms must be confirmed directly with the supplier and backed by contract documentation.
- Grades are indicative and can vary by operation, product specification and shipment. Buyer-specific testing is required before selecting a material for a defined downstream process.
- UK geology references include a 2013 BGS report and an older regional report. Their prospectivity observations should be updated by current data and fieldwork before any investment decision.

End of memo | Hydrogrid PLC | Internal use only